



NOSQL

Unit 1

Subject Code: CIAEC49

B.E Sem: IV

Credits : 1

Why do we need CAP?

- Modern applications require scalability and high availability.
- Data is no longer stored on a single machine; it is distributed across multiple nodes (clusters).

The Challenge: How do we keep data consistent and available when the network between these nodes fails?

What is the CAP Theorem?

Proposed by Eric Brewer in 2000.

Definition: It is impossible for a distributed software system to simultaneously provide more than two out of three guarantees:

- Consistency
- Availability
- Partition Tolerance



The Three Pillars (C-A-P)

- **Consistency (C):** Every read receives the most recent write or an error. All nodes see the same data at the same time.
- **Availability (A):** Every request receives a (non-error) response, without the guarantee that it contains the most recent write.
- **Partition Tolerance (P):** The system continues to operate despite an arbitrary number of messages being dropped (or delayed) by the network between nodes.



The "Choose Two" Rule

- In a perfect world (no network failure), you can have all three.
- In the real world, network partitions (failures) will happen.
- **The Reality:** Since Partition Tolerance (P) is a requirement for distributed systems, you must choose between Consistency (C) or Availability (A).
- Formula: $P + (C \text{ or } A)$

Understanding CP- Reliability over Accessibility

- If a partition occurs, the system shuts down the non-consistent nodes until the partition is resolved.
- **Behavior:** The system returns an error if it cannot guarantee the data is up-to-date.
- **Use Case:** Financial transactions, banking systems, stock trading.
- **NoSQL Examples:** MongoDB, Redis, HBase.

Understanding AP-Accessibility over Reliability

- If a partition occurs, all nodes remain online and accept writes/reads, even if they can't talk to each other.
- **Behavior:** Data might be "stale" (old), but the system stays up.
- **Strategy:** Uses Eventual Consistency to sync data once the partition heals.
- **Use Case:** Social media feeds, shopping carts, YouTube comments.
- **NoSQL Examples:** Cassandra, CouchDB, DynamoDB.